IPL Analytics Dashboard Using Power BI

1. Executive Summary

The IPL Analytics Dashboard is a comprehensive Business Intelligence solution developed using Power BI to analyze historical Indian Premier League (IPL) data. The project leverages match-level and ball-by-ball datasets to provide insights into team performance, batting excellence, bowling effectiveness, venue trends, and match outcomes.

The dashboard was designed with three dedicated analytical pages: Executive Overview, Batting Analytics, and Bowling Analytics. Interactive slicers, dynamic titles, KPI cards, advanced DAX calculations, and custom visualizations were implemented to enhance user experience and support data-driven decision-making.

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2. Business Problem Statement

IPL generates a massive amount of cricket data every season. However, raw datasets do not provide meaningful insights without proper analysis.

The objective of this project was to transform raw IPL data into an interactive analytics platform capable of answering questions such as:

- Which teams have been most successful in IPL history?
- Who are the leading run scorers and wicket takers?
- How does batting performance vary across innings phases?
- Which bowlers create the highest pressure through dot balls?
- What impact does toss decision have on match outcomes?
- Which venues host the most IPL matches?

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3. Dataset Description

The project uses four datasets.

Fact_Match

This table contains match-level information.

Columns include:

- Match Number
- Match Date
- Team 1
- Team 2
- Winner
- Toss Winner
- Toss Decision
- Venue
- City
- Result
- Player of Match
- Season

Purpose:
Provides match-level analysis and executive insights.

Fact_BallbyBall

This table contains delivery-level information.

Columns include:

- Match ID
- Innings
- Over Number
- Ball Number
- Batter
- Bowler
- Batsman Runs
- Extras
- Total Runs
- Wicket Indicator

Purpose:
Used for batting and bowling analytics.

Dim_Player

Contains player information including:

- Player Name
- Batting Name
- Fielding Name

Purpose:
Player-level analysis and filtering.

Dim_Team

Contains team information including:

- Team Name

Purpose:
Team-based analysis and reporting.

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4. Data Cleaning & Transformation

Power Query was used to clean and prepare the data.

Activities performed:

- Removed unnecessary columns
- Standardized column names
- Created Season column
- Removed duplicate records
- Optimized data types
- Reduced model size for improved performance

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5. Data Modeling

A Star Schema model was implemented.

Relationship:

Fact_Match (1) → Fact_BallbyBall (*)

Benefits:

- Faster query performance
- Better DAX calculations
- Simplified dashboard development

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6. DAX Measures Created

Major DAX calculations include:

Executive KPIs

- Total Matches
- Total Runs
- Total Wickets
- Total Sixes
- Total Fours
- Average Runs Per Match

Batting KPIs

- Orange Cap Holder
- Orange Cap Runs
- Total Boundaries
- Total Batters

Bowling KPIs

- Purple Cap Holder
- Purple Cap Wickets
- Dot Balls
- Economy Rate
- Average Wickets Per Match

Dynamic Measures

- Dynamic Dashboard Titles
- Season-based Filtering
- Player-based Filtering

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7. Dashboard Pages

Page 1 – Executive Overview

Purpose:
Provide a high-level summary of IPL performance.

KPIs:

- Total Matches
- Total Runs
- Total Wickets
- Total Sixes
- Total Fours
- Average Score

Visuals:

- Toss Decision Analysis
- Most Successful IPL Teams
- Matches Hosted by Venue
- Top IPL Host Cities
- Team Performance by Season

Insights:

- Mumbai Indians and Chennai Super Kings emerged as the most successful franchises.
- Field-first decisions dominate toss strategies.
- Certain venues consistently host the majority of IPL matches.

Page 2 – Batting Analytics

Purpose:
Analyze batting performance and identify elite batsmen.

KPIs:

- Total Runs
- Total Batters
- Total Boundaries
- Total Sixes
- Orange Cap Holder
- Orange Cap Runs

Visuals:

Orange Cap Leaders

Displays top run scorers in IPL history.

Boundary Kings

Highlights players with the highest boundary count.

Runs Distribution by Overs Phase

Analyzes scoring patterns across:

- Powerplay
- Middle Overs
- Death Overs

Key Findings:

- Virat Kohli leads IPL run-scoring records.
- Boundary scoring contributes significantly to overall batting success.
- Most runs are accumulated during middle overs.

Page 3 – Bowling Analytics

Purpose:
Evaluate bowling effectiveness.

KPIs:

- Total Wickets
- Total Bowlers
- Dot Balls
- Average Wickets per Match
- Purple Cap Holder
- Purple Cap Wickets

Visuals:

Purple Cap Leaders

Top wicket-taking bowlers.

Dot Ball Specialists

Bowlers creating pressure through dot deliveries.

Economy Rate Leaders

Most economical bowlers.

Wickets by Bowling Phase

Analysis of wickets across:

- Powerplay
- Middle Overs
- Death Overs

Key Findings:

- Yuzvendra Chahal emerged as the highest wicket-taker.
- Dot-ball pressure strongly correlates with bowling effectiveness.
- Death overs contribute significantly to wicket-taking opportunities.

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8. Key Insights

Batting Insights:

- Virat Kohli dominates run-scoring statistics.
- Boundary scoring remains a critical contributor to batting performance.
- Middle overs generate the highest run volume.

Bowling Insights:

- Yuzvendra Chahal leads wicket-taking records.
- Dot-ball specialists create significant pressure despite not always leading wicket charts.
- Economy rate provides a more complete measure of bowling effectiveness.

Team Insights:

- Mumbai Indians and Chennai Super Kings remain the most successful IPL teams.
- Venue and toss decisions influence match outcomes.

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9. Challenges Faced

- Handling large ball-by-ball datasets
- Building dynamic DAX measures
- Creating meaningful cricket-specific KPIs
- Optimizing dashboard layout for readability
- Implementing consistent theme design across multiple pages

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10. Future Enhancements

Potential future improvements:

- Player comparison dashboard
- Team comparison dashboard
- Venue performance analytics
- Win probability modeling
- Predictive analytics using machine learning

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11. Conclusion

The IPL Analytics Dashboard successfully converts raw cricket data into a meaningful decision-support system. Through interactive visualizations, advanced DAX calculations, KPI-driven reporting, and strong data modeling practices, the project demonstrates practical Business Intelligence capabilities using Power BI.

The project showcases expertise in data transformation, data modeling, DAX development, dashboard design, and analytical storytelling, making it suitable for portfolio presentation, GitHub publication, and data analyst interviews.